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My research focuses on robust representation learning by designing learnable, parameterised convolutional operators grounded in signal-processing and perceptual principles to improve generalisation under real-world distribution shifts. Application domains are treated as diagnostic tools for studying representation learning, rather than as ends in themselves.

RESEARCH EXPERTISE

- Representation Learning.
- Robustness and Domain Shift.
- Multimodal and Multitemporal Fusion.
- Dataset Creation and Annotation Pipelines.

RESEARCH CONTRIBUTIONS

Physically and Perceptually Grounded Representation Learning

Designed novel convolutional operators embedding learnable signal-processing parameterisations and perceptual principles to improve robustness and generalisation in semantic segmentation under challenging visual conditions.

- **PerceptiveNet.** *CVPR EarthVision 2025 (single-author, spotlight)*. Introduced a learnable Log-Gabor-parameterised convolutional layer enforcing zero DC response, structured frequency coverage, and spatial localisation, yielding improved robustness to occlusion, shadows, clutter, and texture bias in complex, dense, high-resolution scenes.
- **Physics-Aware Water Segmentation.** *IGARSS 2023 (first-author, oral); Elsevier ISPRS (in preparation, single-author)*. Conducted UAV and ground data collection, annotation, and created architectures embedding physically meaningful constraints to address specular reflection, illumination variability, and occlusion in complex high-variability scenes.

Physics-Informed Multimodal and Multitemporal Data Fusion

Developed structured fusion architectures that explicitly model differences in sensing physics across modalities to enable robust multispectral and thermal learning.

- **FusionNet.** *IEEE JSTARS 2026 (single-author); IGARSS 2025 (first-author, oral)*. Introduced a physics-aware mid-level fusion architecture with learnable signal-processing-parameterised convolutional layers, enabling modality-specific inductive biases and robust multispectral and thermal representation learning using curated multimodal datasets.
- **Industrial Asset Detection with EO Data.** *IGARSS 2022 (co-author, oral)*. Applied physics-informed deep learning to detect and characterise pollutant industrial assets using multitemporal Landsat-8 thermal and SWIR data.

Robustness Under Real-World Distribution Shift

Established seasonal and environmental domain shift as a first-class methodological challenge and demonstrated how architectural inductive biases improve robustness and reduce spurious correlations.

- **Structured Convolutions for Robust Domain Generalisation.** *CVPR EarthVision 2023 (first-author)*. Proposed an architecture with learnable signal-processing parameterised convolutional layers and wider receptive fields, achieving improved robustness and reduced spurious correlation sensitivity under domain shift.
- **Robust Learning with Inductive Biases.** *IGARSS 2022 (first-author)*. Showed that learnable, signal-processing parameterised convolutional layers remain robust under severe seasonal domain shift, unlike standard CNNs.

Dataset and Tooling Development for Complex, High-Variability Imagery

Created dataset and annotation pipelines enabling reproducible evaluation of robustness under real-world visual variability.

- Released annotated datasets for complex, high-variability imagery across multispectral, thermal, and multitemporal modalities, capturing occlusion, clutter, and illumination variability.
- Developed a human-in-the-loop annotation pipeline combining foundation-model pre-segmentation with expert refinement, reducing annotation cost while preserving data quality.
- Created pre- and post-processing pipelines for large-scale, high-resolution data, enabling reproducible deployment.

PUBLICATIONS

Voulgaris, 2026. FusionNet: Multi-Spectral and Thermal Deep Learning Fusion for Cement Plant Detection Using a Physics-Aware Geological Ratio. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS). (single-author; Impact Factor 6.3)

Voulgaris, Cole, Crofts, & Sheldon, 2026. Cross-Validated Tree Species Map of A Mixed-Deciduous Woodland Generated Using UAV-Derived RGB-Imagery and Convolutional Neural Networks. Nature Scientific Data. (Under Review)

Voulgaris, 2025. Bridging Classical and Modern Computer Vision: PerceptiveNet for Tree Crown Semantic Segmentation. CVPR Computer Vision and Pattern Recognition Conference, EarthVision. (Spotlight; single-author)

Voulgaris, Bayaraa, & Rossi, 2025. Detecting Cement Plants with Landsat-8: A Physics-Informed, Multi-Temporal, and Multi-Spectral Deep Learning Fusion Approach. IGARSS IEEE International Geoscience and Remote Sensing Symposium. (Oral)

Voulgaris, Philippides, Dolley, Reffin, Marshall, & Quadrianto, 2023. Seasonal Domain Shift Dataset in the Global South: A Deep Features Scene Analysis. CVPR Computer Vision and Pattern Recognition Conference, EarthVision.

Voulgaris, Philippides, & Quadrianto, 2023. Water Physics Aware Semantic Segmentation Through Texture-Biased U-Net Architectures. IGARSS IEEE International Geoscience and Remote Sensing Symposium. (Oral)

Voulgaris, Philippides, & Quadrianto, 2022. Deep Learning Robustness to Domain Shifts During Seasonal Variations. IGARSS IEEE International Geoscience and Remote Sensing Symposium.

Rossi, Tkachenko, Bayaraa, Foster, Reece, Scott, Voulgaris, Christiaen, & McCarten, 2022. Detection and Characterisation of Pollutant Assets with AI and EO to Prioritise Green Investments: The Geoasset Framework. IGARSS IEEE International Geoscience and Remote Sensing Symposium. (Oral)

INVITED TALKS

- **Invited Speaker, Oxford Mathematical Institute Machine Learning & Data Science Seminar, 2025**
Learnable Signal-Processing-Parameterised Convolutional Layers for Robustness Under Real-World Visual Variability.
- **Invited Speaker, Greeks in AI Symposium, 2025**
PerceptiveNet: Bridging Classical and Modern Computer Vision.

RESEARCH LEADERSHIP & COLLABORATIONS

- **AI Lead, British Academy & DEFRA-funded Interdisciplinary Project (with Sussex Science Policy Research Unit, SPRU)**
Developed deep learning methods for land-use classification and multi-domain, multi-temporal aerial data integration. Contributed to a web-based visualisation platform enabling policy and planning in urban sustainability and food security.
- **AI Research Lead, Satellite Applications Catapult (Research Internship, 2022)**
Led the development of large-scale satellite image analysis methods for environmental risk assessment, coordinating model design and data workflows across multispectral and thermal modalities.

ACADEMIC APPOINTMENTS

Postdoctoral Researcher, Department of Biology, University of Oxford (2023 – Present)

- Developed novel deep learning models (PerceptiveNet, FusionNet) for semantic segmentation and multimodal fusion in complex, dense data, improving robustness to occlusion, illumination variations, and other challenging visual conditions.
- Created high-resolution pre- and post-processing data pipelines enabling deep learning models to train reliably on large, heterogeneous imagery and to deploy unlabelled predictions for expert analysis in downstream field workflows.
- Designed and procured shared GPU infrastructure, negotiating a cost-sharing arrangement across two research groups to support deep learning research.
- Delivered technical seminars on deep learning, multimodal data integration, and segmentation methods to researchers across the Department of Biology.

PROFESSIONAL SERVICE

- **Technical Committee Member and Reviewer**, CVPR EarthVision (2024 - 2026)
 - **Co-Organiser**, ICIP Workshop on Computer Vision for Ecological and Biodiversity Monitoring (2025)
 - **Session Chair**, IGARSS - Image Analysis for Remote Sensing of Water Bodies (2023)
 - **Panel Member**, Schmidt AI in Science Programme (Mock Interviews) (2023)
- Provided expert evaluation in deep learning and artificial intelligence for the Department of Biology candidates.

TEACHING

- **Teaching Assistant, Computer Vision (Undergraduate), University of Sussex, 2020**
Coursework marking, lab demonstrations, and support with implementation and theory.
- **Teaching Assistant, Automation and Mechatronics (MSc), University of Sussex, 2019**
Lab demonstrations and support with implementation and theory, contextualised in industry experience across mechatronics and control systems.
- **Teaching Assistant, Industrial Automation Systems (Undergraduate), University of Sussex, 2019**
Lab demonstrations and support with implementation and theory, contextualised in industrial automation and safety-critical systems.
- **Teaching Assistant, Advanced Electronic Systems (MSc), University of Sussex, 2018**
Lab documentation, equipment setup, lab demonstrations, and support with implementation and theory.

EDUCATION

- **PhD in Informatics (EPSRC funded), University of Sussex (October 2018 – August 2023)**
Deep Learning Aerial Scene Analysis: Extracting Salient Features for Domain Adaptation and Semantic Segmentation.
- **MSc Modern Digital Communication Systems, University of Sussex (September 2005 – September 2006)**
- **BEng Hons Electronics with Communication Engineering, University of Brighton (September 2001 – July 2005)**

AWARDS AND FELLOWSHIPS

- EPSRC Doctoral Training Studentship.
- DISCnet Doctoral Training Scholarship.
- Best Research Proposal Award, Machine Learning Summer School (MLSS-Indonesia).
- G-Research Grant (£2,000), 2026, awarded in support of FusionNet (IEEE JSTARS, single-author).

ADDITIONAL EXPERIENCE (SELECTED)

- **Research Intern, Satellite Applications Catapult, 2022**
Developed physics-aware deep learning models for industrial remote sensing, using thermal and spectral features to detect and characterise cement production facilities. Contributed to publicly released datasets and IGARSS 2022 publication.
- **Design Engineer, Oil and Gas Control Systems, Rockwell Automation, 2006–2013**
Designed and implemented sensor-driven control systems in safety-critical Oil and Gas environments. Developed control architectures and software for industrial automation and monitoring systems. Led and mentored junior engineers, supporting project delivery and technical development. Contributed across the full system lifecycle, including functional safety, testing, and deployment under operational constraints.

REFERENCES

Will be given upon request.